Technical Memorandum

Impacts to Hurricane Storm Surge Inundation Resulting from Sea Level Change

Norfolk SLOSH Basin

Responses to Climate Change Pilot Study



***Prepared by*:**

U.S. Army Corps of Engineers
National Planning Center of Expertise for
Coastal Storm Risk Management
National Hurricane Program Office
North Atlantic Division/Baltimore District

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TABLE OF CONTENTS

1.0 INTRODUCTION....1

1.1 Purpose....1
1.2 Overview of Sea Level Change....1
1.3 Overview of the Investigation Team....2

2.0 THE STUDY PROCESS....3

2.1 Study Area and Model....3
2.2 Selection of Sea Level Change Scenarios....4
2.3 Overview of Approach #1 – Bathtub Method....4
2.4 Overview of Approach #2 – Model Method....4

3.0 RESULTS....5

4.0 CONCLUSIONS AND RECOMMENDATIONS....9

5.0 REFERENCES....11

6.0 ACRONYMS....12

TABLES:

Table 1: Sea Level Change Scenarios.... 4

Table 2: Average Increase in Storm Surge Due to Sea Level Change Over Norfolk Basin.... 5

Table 3: Average Increase in Storm Surge Due to Sea Level Change Over Land Within Norfolk Basin.... 6

FIGURES:

Figure 1: Norfolk SLOSH Basin Grid Area.... 3

Figure 2: Deviation from Average Storm Surge Increase for a Category 1 Storm with a Sea Level Change Increase of 0.53 Feet (2040 Low) within the Norfolk Basin.... 7

Figure 3: Deviation from Average Storm Surge Increase for a Category 4 Storm with a Sea Level Change Increase of 5.52 Feet (2100 High) within the Norfolk Basin.... 8

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1.0 INTRODUCTION

1.1 Purpose

This investigation focuses on an assessment of the impact of Sea Level Change (SLC) on the severity of hurricane storm surge inundation within the Norfolk SLOSH (Sea, Lake, and Overland Surge from Hurricanes) Basin. Flooding due to storm surge inundation is widely considered to be the most likely hurricane hazard to cause mass casualties and poses a major threat to critical infrastructure. Identification of the areas that are subject to storm surge inundation is a key component of hurricane readiness, evacuation, and response planning. Maps of storm surge inundation risk are used as the basis for Hurricane Evacuation Studies (HESs), for determining locations of hurricane shelters, and are an important factor in the management of critical facilities and infrastructure. Storm surge inundation risk area maps can be used to determine which critical facilities and infrastructure (such as schools, police stations, fire stations, bridges, emergency operations centers, etc.) may be in a surge risk area, and if relocation or mitigation of this infrastructure should be considered. The maps are also an important resource to be used when making decisions on where to build new infrastructure.

Current hurricane planning, evacuation studies, and other important critical facility decision-making efforts reference surge inundation risk maps that are based on current sea level conditions. This investigation will provide planners and decision makers with an initial assessment of two methods to consider when analyzing the possible impact of SLC on storm surge inundation risk. The first approach is referred to as the "bathtub" method, which is the process of adding SLC amounts to known current conditions hurricane storm surge heights to arrive at future conditions surge heights. The second approach is referred to as the "model" method: the process of modeling surge from hurricane events based on increased starting water levels resulting from predicted SLC. The bathtub method is a much simpler, quicker, and less expensive method. This investigation provides the initial data that will be needed to support future investigations to determine in what conditions/scenarios the bathtub method may be acceptable, and what conditions/scenarios the model method would be required.

1.2 Overview of Sea Level Change

Geologic evidence suggests global mean sea level (GMSL) has fallen and risen with minimums and maximums occurring during cold glacial and inter-glacial warm periods respectively. At the peak of the most recent glacial period about 20,000 years ago, GMSL is inferred to have been on the order of 100-120 meters lower than at present (NRC 1987, IPCC 2007). During the last inter-glacial period, about 125,000 years ago, sea level was four to six meters higher than at present. The earth entered the present inter-glacial warm period following the peak of the last Ice Age about 12,000 years ago (CCSP 2009). After a rapid initial rise, GMSL is interpreted as having stabilized within a meter or so of its present value over the last several thousand years (NRC 1987, IPCC 2007). The Intergovernmental Panel on Climate Change (IPCC) (2007) concludes that GMSL rose at an average rate of about 1.7 ± 0.5 mm/year during the twentieth century.

Recent climate research has documented global warming during the 20th Century, and has predicted either continued or accelerated global warming for the 21st Century and possibly beyond (IPCC 2007). One impact of continued or accelerated climate warming is continued or accelerated rise of GMSL (USACE 2011).

SLC can cause a number of impacts in coastal and estuarine zones, including changes in shoreline erosion, inundation or exposure of low-lying coastal areas, changes in storm and flood damages, shifts in extent and distribution of wetlands and other coastal habitats, changes to groundwater levels, and alterations to salinity intrusion into estuaries and groundwater systems (CCSP 2009). Climate change may also alter the frequency and severity of tropical storms which could secondarily influence sea level (IPCC 2007).

1.3 Overview of the Investigation Team

The U.S. Army Corps of Engineers (USACE), Baltimore District (NAB), representing the North Atlantic Division (NAD) as the Hurricane Program Office for the Planning Center of Expertise (PCX) for Coastal Storm Risk Management, was the lead for this investigation. The Hurricane Program Office has been involved in the production, management, and general oversight of the storm surge inundation mapping projects and HESs on a national scale. These studies are funded through the National Hurricane Program (NHP), a multi-agency partnership and collaborative effort between the Federal Emergency Management Agency (FEMA), National Oceanic and Atmospheric Administration (NOAA), USACE, and others. The products of the NHP are provided to State and Local governments for use in hurricane preparedness, planning, and hurricane emergency management decision making.

NAB leveraged the experienced team at the NOAA National Hurricane Center (NHC), who produced the storm surge modeling output data needed for this investigation. The NHC used the SLOSH model to produce the data. SLOSH is the official model used by the NHC to support hurricane storm surge forecasting during approaching storms and to identify areas that have storm surge inundation risk in support of hurricane readiness, evacuation, and response planning efforts.

2.0 THE STUDY PROCESS

2.1 Study Area and Model

The study area for this pilot study is the most recent version of the Norfolk SLOSH Basin (also called ord3, output elevations referenced to NAVD88) (NOAA 2012). The SLOSH model is the official model used by the NHC to forecast and predict impacts from hurricane storm surge. The SLOSH model "Maximum of Maximums" (MOM) simulation process is used to support planning and determines all areas that have at least some risk of surge inundation from a hurricane within a given Category on the Saffir-Simpson Hurricane Wind Scale. The MOM output elevations also establish maximum potential storm tide (surge + tide) heights per category within each basin grid cell. The MOM output is produced by simulating thousands of hurricanes with different tracks, angles of approach, radius of maximum winds, and forward speeds within each category, and recording the highest storm tide elevation recorded in each grid cell regardless of which storm simulation produced it. A depiction of the Norfolk SLOSH Basin is shown in Figure 1 (NOAA 2014). For more information on the SLOSH model and the MOM product, please visit http://www.nhc.noaa.gov/surge/slosh.php.

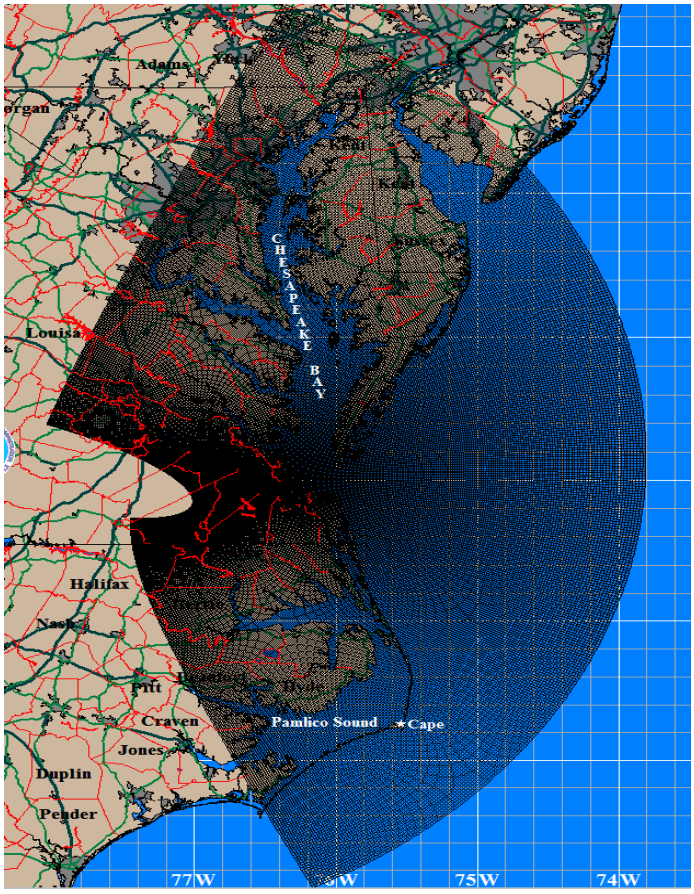


Figure 1: Norfolk SLOSH Basin Grid Area

2.2 Selection of Sea Level Change Scenarios

Using USACE Engineer Circular (EC) 1165-2-212, titled Sea-Level Change (SLC) Considerations for Civil Works Programs (USACE 2011), SLC scenarios were selected and increase amounts established. SLC levels were calculated for 6 locations within the Norfolk Basin and then averaged. The scenarios and increase amounts are shown in Table 1. These scenarios were selected to establish a range of values that would incorporate low, intermediate, and high scenarios and also evaluate shorter range and longer range SLC.

Table 1: Sea Level Change Scenarios

Sea Level Change Scenario	Value (Feet)
2040 Low	0.53
2040 Intermediate	0.73
2040 High	1.38
2100 Low	1.19
2100 Intermediate	2.23
2100 High	5.52

2.3 Overview of Approach #1 – Bathtub Method

This approach takes the pre-existing Norfolk SLOSH Basin MOM simulation output storm tide elevations (high tide), based on current conditions, and adds the six different sea level change amounts. This results in storm surge + SLC future conditions water levels for each SLC scenario. This is a simplified approach that can be accomplished quickly and at a low cost.

2.4 Overview of Approach #2 – Model Method

This analysis requires the actual modeling of hurricane surge using the calculated SLC amounts added to the starting water level. Unlike the bathtub method, the model approach calculates the impact of SLC on hurricane storm surge based on a model run by the NHC. For this investigation, the NHC reproduced the Norfolk Basin SLOSH MOM product (high tide) based on each of the six provided SLC scenarios. This involved running thousands of hurricane scenarios in the SLOSH model for each hurricane category, and recording the highest calculated water level in each grid cell. The resulting MOM + SLC SLOSH outputs allow for a grid cell to grid cell comparison with the bathtub method calculations. Nothing was changed in the model and basin that was used to produce the current conditions Norfolk Basin SLOSH MOM output other than the SLC amounts being added to the starting water levels. This includes no geomorphology changes that would predict or account for future erosion, accretion, land use, etc.

3.0 RESULTS

The average increase in storm surge for each level of modeled SLC shown in Table 1 was calculated for all Norfolk SLOSH Basin cells with inundation values for the entire basin, including those over open water. Only cells with inundation values in both the specific SLC SLOSH run and also in the current 2012 SLOSH run without SLC modeled were included in the analysis. The results are shown in Table 2.

Table 2: Average Increase in Storm Tide Due to Sea Level Change Over Norfolk Basin

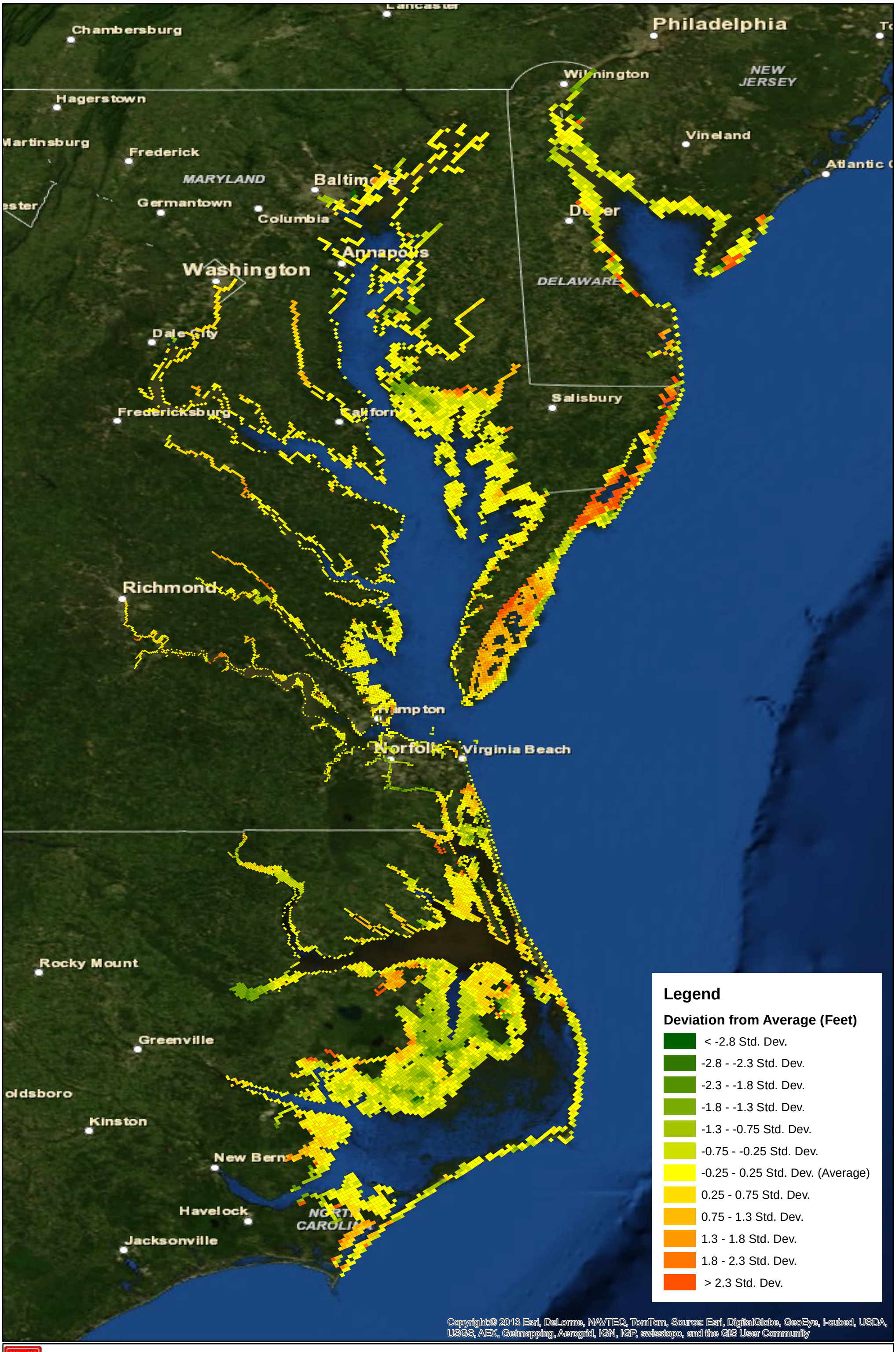
BATHTUB METHOD	**MODEL METHOD**			
Storm Surge Increase over Current Conditions	**Average Increase in Storm Tide Compared to Current Conditions (Feet)**			
Saffir-Simpson Hurricane Wind Scale	**Category 1**	**Category 2**	**Category 3**	**Category 4**
Increase 0.53 Feet (2040 Low)	0.50	0.49	0.49	0.50
Increase 0.73 Feet (2040 Intermediate)	0.69	0.68	0.68	0.69
Increase 1.19 Feet (2100 Low)	1.18	1.16	1.16	1.17
Increase 1.38 Feet (2040 High)	1.38	1.36	1.19	1.36
Increase 2.23 Feet (2100 Intermediate)	2.16	2.11	2.10	2.09
Increase 5.52 Feet (2100 High)	5.46	5.42	5.44	5.50

The average increase in storm surge for each level of modeled SLC was calculated again for all Norfolk SLOSH Basin cells with inundation values in both the specific SLC SLOSH run and also in the current conditions 2012 SLOSH run without SLC modeled, but this time only for areas over land. SLC over open water areas is not as important to planners and decision makers as over land; therefore, open water cells were removed from the analysis. A USACE NAD water shoreline ArcGIS shapefile was used in order to identify those cells that cover land. Results are shown in Table 3.

Table 3: Average Increase in Storm Tide Due to Sea Level Change Over Land Within Norfolk Basin

BATHTUB METHOD Storm Surge Increase over Current Conditions	MODEL METHOD Average Increase in Storm Tide Compared to Current Conditions (Feet)			
Saffir-Simpson Hurricane Wind Scale	**Category 1**	**Category 2**	**Category 3**	**Category 4**
Increase 0.53 Feet (2040 Low)	0.52	0.50	0.52	0.53
Increase 0.73 Feet (2040 Intermediate)	0.71	0.70	0.71	0.73
Increase 1.19 Feet (2100 Low)	1.22	1.18	1.21	1.24
Increase 1.38 Feet (2040 High)	1.43	1.38	1.38	1.45
Increase 2.23 Feet (2100 Intermediate)	2.23	2.14	2.16	2.16
Increase 5.52 Feet (2100 High)	5.65	5.72	5.86	5.99

Figures 2 and 3 depict the deviation for each SLOSH grid cell from the average increase that is shown in Table 3 (Model Method) for a SLC increase of 0.53 feet for a Category 1 storm and also for a SLC increase of 5.52 feet for a Category 4 storm. It was noted that while most of the SLOSH cells had an increase that was close to the basin-wide average, there were areas where storm tide increase was substantially lower or higher than the average modeled water level. The figures also show where a substantial difference exists between the "bathtub method" storm tide water levels and the "model method" storm tide water levels. Cells where the modeled storm tide value was significantly different from the average are also significantly different from the bathtub method value for that cell.



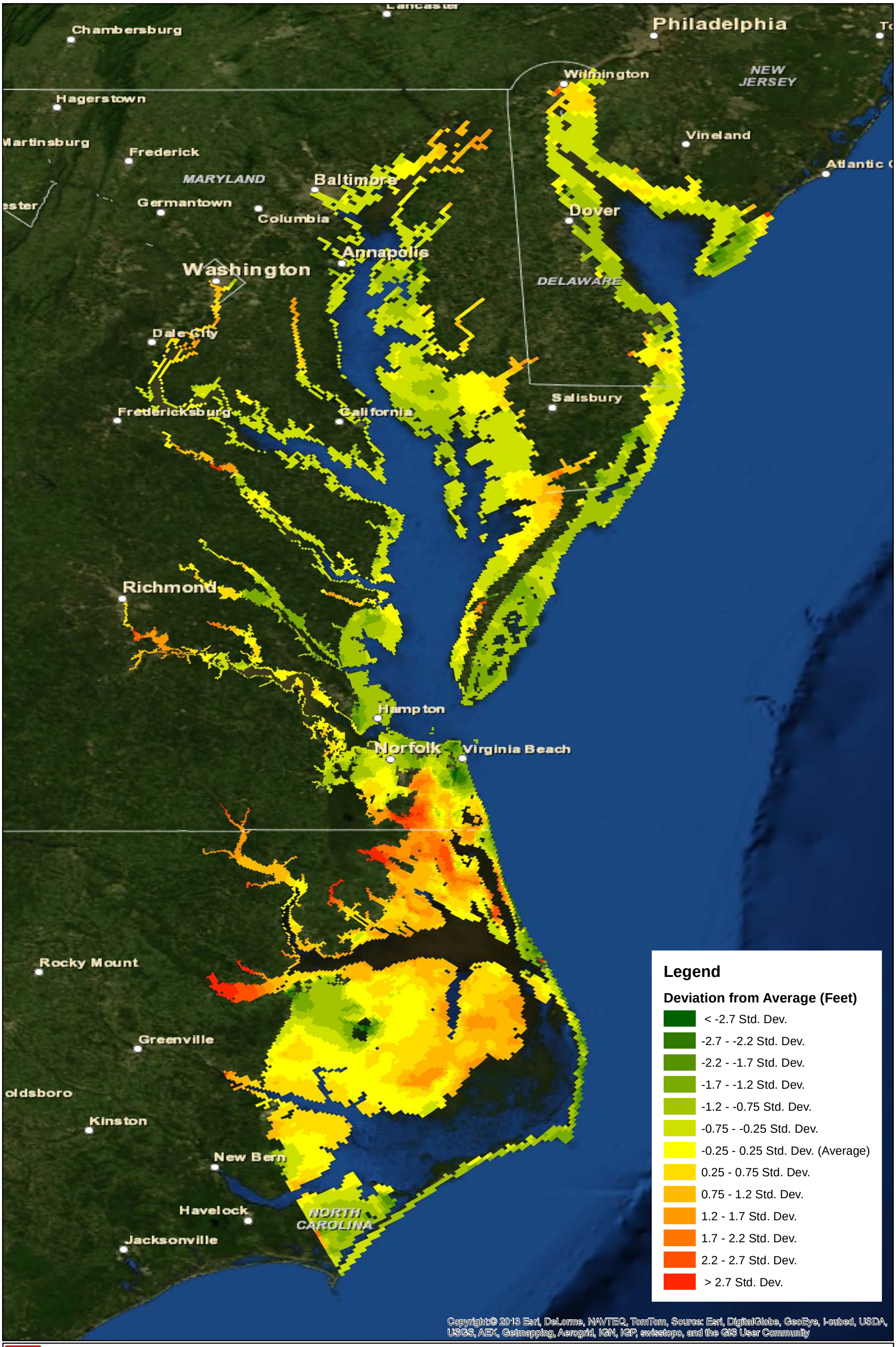
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Figure 2: Deviation from Average Storm Surge Increase for a Category 1 Storm with a Sea Level Change Increase of 0.53 Feet (2040 Low) within the Norfolk Basin

0 12.5 25 50 Miles

1 inch = 25 miles

N W E S



US Army Corps of Engineers
Baltimore District

Figure 3: Deviation from Average Storm Surge Increase for a Category 4 Storm with a Sea Level Change Increase of 5.52 Feet (2100 High) within the Norfolk Basin

0 12.5 25 50 Miles

1 inch = 25 miles

N W E S

4.0 CONCLUSIONS AND RECOMMENDATIONS

This investigation provided the data needed to analyze two methods to consider when accounting for the possible impact of storm surge inundation based on various SLC scenarios within the Norfolk Basin: 1) the bathtub method and 2) the model method using the SLOSH model.

It should be noted that the main purpose of this investigation was to produce the modeling data needed to make comparisons between the bathtub method and the model method. Once those calculations were complete and the data available, a limited statistical analysis was conducted to compare the results using very simplified methods. Although there are some general initial conclusions discussed in Section 3, these are based on the simplified initial look at the model results. There are additional analyses that would need to be conducted before any confident assumptions can be made in comparing the bathtub method to the model method; however, they were outside the scope of this investigation. These are further discussed below.

Figures 2 and 3 show that there could be a wide deviation from the average storm surge increase for a given SLC value that is run through the SLOSH model, although most of the cells are close to the average. There is a greater deviation from the average with increased SLC value and storm category. There are a variety of possible factors for these "outliers", including, but not limited to: topography of the continental shelf and inland areas, shape of the coastline, and resolution of the SLOSH grid cell. Further QA/QC of the model results, especially in areas where the modeled surge increase/decrease was significantly higher than average, is needed to verify accuracy of the results. Once result accuracy has been verified, further investigation would be needed to identify specific conditions that may have led to the greater than average increase or decrease. If these conditions could be identified, they would become an important consideration when determining whether to use the bathtub method or the model method. The bathtub method would be overestimating or underestimating in these cases. Investigation the cause of these "outliers" was not a part of this study.

SLOSH outputs above a vertical datum (NAVD88) were produced during this study. In future endeavors, SLOSH output depths above ground could be analyzed. This method would include more inundated grid cells over land since only cells with inundation values in both the specific SLC SLOSH runs and also in the current conditions SLOSH data could be used in this analysis to compare calculated surge increases resulting from the two methods. This would allow for an analysis of the change in inland extent of inundation for the various SLC scenarios compared to current conditions inland extent of inundation for each hurricane category. This was also outside the scope of this investigation.

The simplified statistical analysis did not account for variation in the size of grid cells. All grid cells were analyzed regardless of how far they are from the center of the basin, resulting in a range of grid cell resolution. Further investigation is needed to determine if the difference between the bathtub method results and model method results varies between higher resolution grid cells closer to the basin center and lower resolution grid cells farther away from the center. This would identify to what extent model resolution impacted the results, if there may be a

minimum model resolution needed to support the model method, and whether the bathtub method may be preferred if only lower resolution modeling data could be produced.

The Norfolk SLOSH basin has certain shoreline topography and bathymetry conditions. This same analysis would need to be repeated in a SLOSH basin that has different shoreline topography and bathymetry conditions before any further conclusions could be drawn establishing when the bathtub method may be acceptable. The New Orleans SLOSH Basin, also known as the Central Gulf Basin, may be a good candidate for the repeat analysis based on different shoreline topography and bathymetry conditions.

In summary, it appears on the surface that in many scenarios the bathtub method could be a reasonable low risk, low cost, and minimal effort approach to estimating the impact of SLC on hurricane storm surge. However, considerable additional analysis is needed to support this assumption, and would be needed before any guidance on what situations and scenarios the bathtub method would be acceptable. This investigation produced the data that is needed for these additional investigations discussed in this section. If the bathtub method were proven to be acceptable in some situations, it would likely reduce time and funding needed on some future feasibility studies, as the cost and time needed for the model method is much greater.

Please note that all data produced for the analysis can be provided by the Baltimore District, including the SLC SLOSH MOM outputs in GIS shapefile format.

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6.0 ACRONYMS

EC	Engineer Circular
FEMA	Federal Emergency Management Agency
GMSL	Global Mean Sea Level
HES	Hurricane Evacuation Study
IPCC	Intergovernmental Panel on Climate Change
MOM	Maximum of Maximums
NAB	Baltimore District
NAD	North Atlantic Division
NHC	National Hurricane Center
NHP	National Hurricane Program
NOAA	National Oceanic and Atmospheric Administration
PCX	Planning Center of Expertise
SLC	Sea Level Change
SLOSH	Sea, Lake, and Overland Surge from Hurricanes
USACE	U.S. Army Corps of Engineers